VISHNU PATLOLLA

Staff QA Engineer · SDET · AI-Powered Quality Architect

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PROFESSIONAL SUMMARY

Staff-level SDET and QA Engineer with 10+ years across gaming, healthcare, financial services, and government. Started as a software developer in 2015 and moved into test automation, and still builds CI/CD pipelines and developer tooling. Sole architect of a production-grade multi-platform test automation monorepo (TypeScript, WebdriverIO, Playwright, Appium) covering 15+ backend microservices, 8 CI/CD pipelines, and a custom ClickHouse observability service. Games-platform QA owner who validates platform features (ad monetization, push notifications, telemetry SDK, GDPR/COPPA compliance) before game-team integration, and who supports developers directly on pipelines, test tooling, and handoff. Built 25+ AI agents and skills with Claude Code that automate test authoring, telemetry validation, and QA reporting across the organization.

AI-DRIVEN QA AND ENGINEERING

- **Claude Code agents and skills (25+):** Built and shared AI agents and skills that automate test authoring, telemetry validation, and QA reporting across the organization.
- **Claude models and MCP integrations:** Connected Claude models to Git, Slack, Singular, ClickHouse, Notion, and Jira through MCP, and set up automated flows across them.
- **Mergemon built with Claude LLMs:** The game was built with Claude LLMs, and its automation was set up by a bot that learns from and fixes issues.
- **Self-fixing E2E:** Set up an automated agent that checks E2E failures and fixes them without manual intervention.
- **Playwright with MCP:** Used the Playwright MCP server with Claude Code to author and debug web E2E tests, so the browser can be driven and inspected from the agent session.
- **Test authoring agent:** Built api-test-writer-v2, an 11-step guided agent that walks engineers from branch creation → spec authoring → PR → CI monitoring and failure triage, with auth-pattern handling and coding-standard enforcement.
- **Telemetry inspection skill:** Queries ClickHouse directly, auto-resolves player IDs and time ranges, runs spec-vs-actual data-quality analysis, cross-references known ClickUp bugs, and exports CSVs. Used across 10+ game projects.
- **Handoff skill:** Turns initiative tickets and Notion/Figma specs into the ClickUp hierarchy the game teams use, so every feature is delivered the same way.
- **Adoption and measurement:** Built meta-skills (growth-scanner, session-review, weekly-qa-ai-updates) to measure and grow AI-tooling adoption across the QA team. Authored the Spire Telemetry QA Test Plan (8 Notion sub-pages), adopted as the Data team's official handoff to game teams.
- **Learning new platforms:** Learned ClickHouse, Singular, AppLovin MAX, and Firebase on the job, and used them to build the telemetry, ad, and purchase checks.
- **Impact:** AI-assisted tools now cover test authoring, telemetry checks, and feature handoff across the QA team and 5+ game teams.

TECHNICAL SKILLS

Languages: TypeScript · JavaScript · Python · Java · C# · .NET MVC

Test Automation: Playwright (incl. MCP) · WebdriverIO · Appium · Cypress · Selenium · Cucumber/BDD · Mocha/Chai · RestAssured · Robot Framework · PyTest · Postman · k6 · Jira · Sauce Labs · Playmode

Development: Lerna/npm-workspaces monorepos · Private npm package publishing (Nexus) · Unity · Docker · Kubernetes

CI/CD and Infrastructure: GitHub Actions · Azure DevOps · Jenkins · Unity Cloud Build · TestFlight / AppCenter · OIDC · Matrix builds

Data, Analytics and Ads: ClickHouse · Singular · AppLovin MAX · Firebase · MongoDB · PostgreSQL · MS SQL · Grafana · Allure · Hex

APIs and Compliance: REST · OAuth2/JWT · OIDC · OpenAPI/Swagger · API backward compatibility testing · OneTrust · GDPR · CCPA · COPPA

AI and Tooling: Claude Code (25+ agents and skills) · Playwright MCP · LLM-assisted test generation · Self-healing automation · AI anomaly detection · Slack Block Kit bots · Claude models · MCP (Git, Slack, Singular, ClickHouse, Notion, Jira) · ClickUp automation

PROFESSIONAL EXPERIENCE

Staff QA Engineer | **Fortis Games** *Feb 2025 – Nov 2026*

Vancouver, BC (Remote) — Platform / Central QA (cen-automation)

Sole QA architect for a cloud-native services platform powering multiple game titles. Owned API test automation, mobile and end-to-end compatibility testing, telemetry pipeline validation, GDPR compliance, production observability, and AI-driven QA tooling across 15+ microservices, gating every platform feature before it reaches 5+ game teams.

Test Automation Framework Architecture

- **Monorepo from zero:** Architected a Lerna/npm-workspaces TypeScript monorepo (6 packages, 83 spec files, 374 tests, ~14.7K LOC) spanning WebdriverIO+Mocha (API), Playwright (web), and Appium+Sauce Labs (mobile) across 15+ platform microservices; 424 commits over 14 months.
- **Playwright with MCP:** Used the Playwright MCP server with Claude Code to author and debug the web E2E suite.
- **API client and test utilities:** Built the core API test framework: a packaged typed API client, a mail client for OTP and email flows, and environment-driven configuration.
- **Reusable SDK packages:** Published @fortis/* packages to a private Nexus registry: a typed Axios API client with Bearer-token handling, a central endpoint registry for 9 services, and 25+ service helper modules.
- **End-to-end telemetry validation:** Built a 1,890-line data-platform E2E spec covering the full event pipeline, from guest creation through Maelstrom, Passport, Prism, and vendor sync (CleverTap and Singular), with polling and direct MongoDB assertions for eventual consistency.
- **Auth optimization:** Designed a headless login flow that caches the JWT per suite run, eliminating per-test login overhead; replaced brittle 35 s sleeps with adaptive 90 s polling.
- **Impact:** One shared framework for 15+ services across API, web, and mobile, instead of separate suites per platform.

API Backward Compatibility Testing

- **Retro Daily E2E:** Runs the production client against dev every day, so backend changes that break shipped clients show up early.
- **Retro RC E2E:** Runs the production client against each staging backend deploy before release.
- **Drift warning:** Posts a Slack warning when staging is not running the latest release backend, and fixed misleading results in the lanes.
- **Release-branch back-ports:** Back-ported the compatibility fixes to the 0.19 and 0.20 release branches.
- **Impact:** Backend changes are checked against the production client daily and on every staging deploy, before players see them.

API Coverage

- **Identity and players:** Forge identity and auth, profiles, Passport (player IDs), OTP.
- **Player lifecycle:** Promotion, player data, status, and deletion.
- **Content:** Facts API, manifest, and snapshot.
- **Compliance:** Agreements (Terms of Service and privacy), age-gate actions, geo-fencing and geolocation.
- **Commerce and messaging:** Marketplace refund states, IAP purchase validation, IGM, mailer end-to-end.
- **General:** Error code validation.
- **Schema validation:** Validated APIs manually in Postman first to understand schemas and payloads, then automated them.

Performance Testing

- **API load testing with k6:** Performance-tested every API covered on the platform.
- **Frame-rate testing on Sauce Labs:** Measured frames per second across different devices to check game performance.

Mobile Automation: Native Ads and IAP on Sauce Labs

- **Nightly ads and IAP check:** WebdriverIO and Appium on a private Android device in Sauce Labs. Covers two rewarded-ad placements with reward verification, and the Buy flow through to Google Play. Slack report uses the E2E format with Sauce video links.
- **Google Play purchase flow:** Automated the purchase flow on the Cen Tech demo app and checked each purchase against the marketplace.
- **Ad unit and device fix:** Found and fixed a dead dev Android ad unit, and registered the test phone as an AppLovin test device.
- **Nightly fail-fast fix:** The nightly now fails fast with the cause when the K-ID age-verification page appears, reports startup failures separately, and only picks builds that still have their APK.
- **Age assurance blocker:** Working with the platform team on the K-ID age-verification change on dev, which currently blocks the nightly run.
- **Findings shared with other teams:** Google Play only sells to Play-installed builds, which shaped the purchase phase 2 plan. consent-QA parsing broke under Mobile Protect, and its R12 rule misread failed purchases. The Firebase 25-parameter limit was dropping analytics fields.
- **Impact:** Ad and purchase problems are caught nightly on a real device, and the findings went to Play Console, consent, and analytics owners.

Platform → Game Team Feature Enablement

- **Ad monetization QA:** Owned AppLovin MAX mediation testing across 4 ad networks (Unity Ads, Meta, AdMob, AppLovin bidders + waterfall), verifying SDK init, per-network fill, server-side toggles, and reward-grant integrity. Defined the acceptance bar (fill >95%, eCPM >$8, load <5 s, crash <0.1%) with a per-network troubleshooting matrix.
- **Push and deep-linking sign-off:** Built the CleverTap production sign-off framework (35 test cases, 8 categories) covering push delivery across app states, deep-link navigation to 8+ in-game destinations, deferred deep-link queueing, and multi-device targeting.
- **Privacy compliance gate:** Owned the COPPA/GDPR compliance gate for push: OneTrust consent toggle, consent and permission matrix, and age-gating (US 13+, EU 16+), blocking release on any compliance failure.
- **Shift-left event validation:** Built a 4-step workflow (ClickHouse delivery → Hex data-quality analysis → 4-hour monitoring → Notion sign-off) that catches schema mismatches, buffering errors, and null fields 2–3 weeks earlier, cutting post-integration event bugs by ~50%.
- **Handoff automation:** Automated the platform-to-game-QA handoff with a Claude Code skill that reads initiative tickets, extracts Notion/Figma specs, generates a control + test-plan + bug-bucket hierarchy in ClickUp, and updates the central tracking doc.
- **Impact:** Every platform feature gates on functional and regulatory criteria before it reaches 5+ game teams, with about half as many post-integration event bugs.

Compatibility Testing: Mergemon

- **Daily E2E reporting:** Redesigned the Slack summary layout. Every failure is reported, including setup crashes.
- **Play-through stability:** Stabilized 0.20 onboarding and tutorial flows (Ember hold, Sanctuary quest entrance, collection gift, Rescuemon merge), and fixed Village overlap recovery and the Residents curtain wait.
- **E2E coverage and infrastructure:** Added coverage for settings legal links, and a Data Import lane for moving bundles between environments.

CI/CD Pipelines and Production Observability

- **8 GitHub Actions pipelines:** API E2E, web E2E, data-platform journey, passport smoke, PROD health, DSAR validation, and ClickHouse monitoring, with OIDC auth, GitHub App tokens, matrix builds, off-peak cron scheduling, and secret validation.
- **API framework CI and reporting:** Workflows run daily or twice daily, with manual inputs. Slack Block Kit reports @-mention owners on failures. Diagnosed and fixed a bug that was miscounting retried tests in CI summaries.
- **Production health checks:** Per-game PROD health workflow (gameId input) that checks production services and logs each URL. Built a 17-spec read-only PROD smoke suite with Slack reporting.
- **ClickHouse anomaly detection:** Built a 955-line monitoring service that runs daily across 8 games (Lotus, Onyx, Parachute, Bishop, Fusebound, Flux, Hyperlight, Octopus). It computes rejection-ratio thresholds over rolling 4-hour windows and posts Slack Block Kit digests to game-specific channels, with central alert routing.

- **Impact:** Developers get pipeline results in Slack with the right owner tagged, and telemetry problems reach the right game channel.

Sprint Process, Releases, and UAT

- **Sprint work in Jira:** Followed the full sprint model day to day in Jira.
- **Release support:** Supported releases through testing, and prepared and signed off the release document.
- **Environments and UAT:** Ran testing across environments and supported user acceptance testing.

Developer Support and Tooling

- **Pipelines and shared packages:** Built the CI workflows that run on engineers' branches, and published reusable SDK packages to a private registry for teams to use.
- **Platform ops Slack bot:** Shipped a Socket Mode Slack bot that classifies platform-engineering requests, auto-creates ClickUp tickets, routes them to the correct POD, and posts interactive daily digests. Remediated SAST findings (path-traversal, Cycode/Datadog).
- **Impact:** Engineers get routed to the right team without manual triage, and test-writing questions are handled by a guided agent.

GDPR / DSAR Compliance Automation

- **Production validation workflow:** Built a PROD DSAR/OneTrust GDPR workflow that creates a real player, triggers a deletion request, and verifies fan-out completion across 5 downstream systems via OneTrust OAuth2 orchestration and adaptive polling. Runs on a weekday CI schedule with a Slack report.
- **Impact:** Deletion requests are verified end to end in production every weekday.

Senior QA Automation Developer | **BC Registries — Government of British Columbia** *Jul 2024 – Dec 2024*

Vancouver, BC

- **Framework development:** Built an end-to-end Cypress + TypeScript automation framework covering API and UI tests for a legacy-to-modern platform migration; extended coverage with Python/Robot Framework suites.
- **Pipeline and performance:** Implemented parallelized Azure DevOps pipelines to cut build times; executed k6 load testing to benchmark throughput and latency under peak loads; generated Grafana/Azure dashboards for stakeholders.
- **AI-augmented testing:** Applied self-healing locators and LLM-assisted test-script generation to increase framework robustness and accelerate delivery.
- **Impact:** Faster pipelines and load-test results that stakeholders could read directly from dashboards.

Lead Software Developer, Quality | **Unity Technologies** *Jan 2021 – Jun 2024*

Vancouver, BC

Led QA across 3 client-facing projects over 3.5 years, spanning game commerce, digital twins, and AR social. Owned automation strategy, mobile CI/CD, release management, and team mentorship, and built the pipelines and test tooling that developers used day to day.

Walmart — Unity Game Commerce Integration (iOS / Android)

- **Custom SDK testing:** Developed an automated test suite for a Unity package that redirected in-game users to Walmart for shopping, targeting both iOS and Android builds via Unity's internal test tooling.
- **API automation:** Built a RestAssured + Java framework to test REST APIs deployed on Google Cloud; configured pipeline-scheduled runs to validate API contracts on a regular interval.
- **QA build enablement:** Worked with developers to create a dedicated QA build using C# defines to skip authentication, enabling test execution on local machines and QA VMs without credential dependencies.
- **CI pipeline:** Set up GitHub Actions to execute Playmode automated tests on every new build and post results to Slack, giving developers feedback on each change.
- **Impact:** Developers got test results on every build, without waiting for a manual QA pass.

YVR Airport — Unity Digital Twin (iOS / Windows)

- **Team lead:** Led a QA team developing automated tests using GameDriver IO for a Unity-based digital twin application running on iOS and Windows devices for Vancouver International Airport.
- **Automation strategy:** Created a comprehensive test plan to replace manual regression execution with automated runs for regular builds.

- **Nightly CI pipeline:** Set up a Jenkins nightly job to execute UI tests on the Windows executable build on a dedicated QA VM, with results distributed via Outlook and Microsoft Teams.
- **Impact:** Full regression suite delivered within the committed timeframe, replacing manual regression runs.

GUM — Augmented Reality Social App (Gen-Z)

- **Mobile UI automation:** Built automated UI tests for the iOS mobile app using Appium and Unity's internal testing package, enabling repeatable regression testing on every release candidate.
- **Release management and pipelines:** Served as Release Manager. Set up CI/CD using Unity Cloud Build and distributed builds via Apple TestFlight and Microsoft AppCenter.
- **Cross-functional collaboration:** Partnered with Designers and User Research teams to integrate early customer feedback into the testing strategy, validating app experience before formal QA cycles.
- **Mentorship:** Mentored interns and guided them to implement tests for new features, including early testing on feature branches before merge, raising overall team velocity and test coverage.
- **Impact:** Each release candidate was tested before QA cycles, and interns shipped tests with their features.

SDET | **Global Relay Communications** *May 2020 – Dec 2020*

Vancouver, BC

- **Microservices testing:** Developed test automation for Kubernetes-deployed Java microservices handling REST, Thrift, and Kafka messaging; executed performance tests and maintained QA environment deployments via Docker/Kubernetes.

Test Automation Developer | **TD Bank** *Feb 2019 – Apr 2020*

Vancouver, BC

- **Full-stack test framework:** Built a Java/RestAssured API + Selenium UI automation framework for the Wealth Data Strategy platform; integrated with Jenkins pipelines and Extent Reports for automated test reporting.

Application Development Analyst | **Cigna Health Insurance** *Jul 2015 – Jan 2018*

Toronto / Hyderabad

- **Application development:** Developed a .NET MVC report-validation web application, which was my start as a software developer.
- **QA and automation:** Built Selenium/Cucumber test suites for the application and demonstrated RPA (Blue Prism) feasibility for insurance workflow automation.

EDUCATION

Bachelor of Technology, Electronics & Communications Engineering *2011 – 2015*

JNTUH, Hyderabad

CERTIFICATIONS

- **Machine Learning**, Stanford University / Coursera (Prof. Andrew Ng)
- **Kubernetes & Docker**, LinkedIn Learning